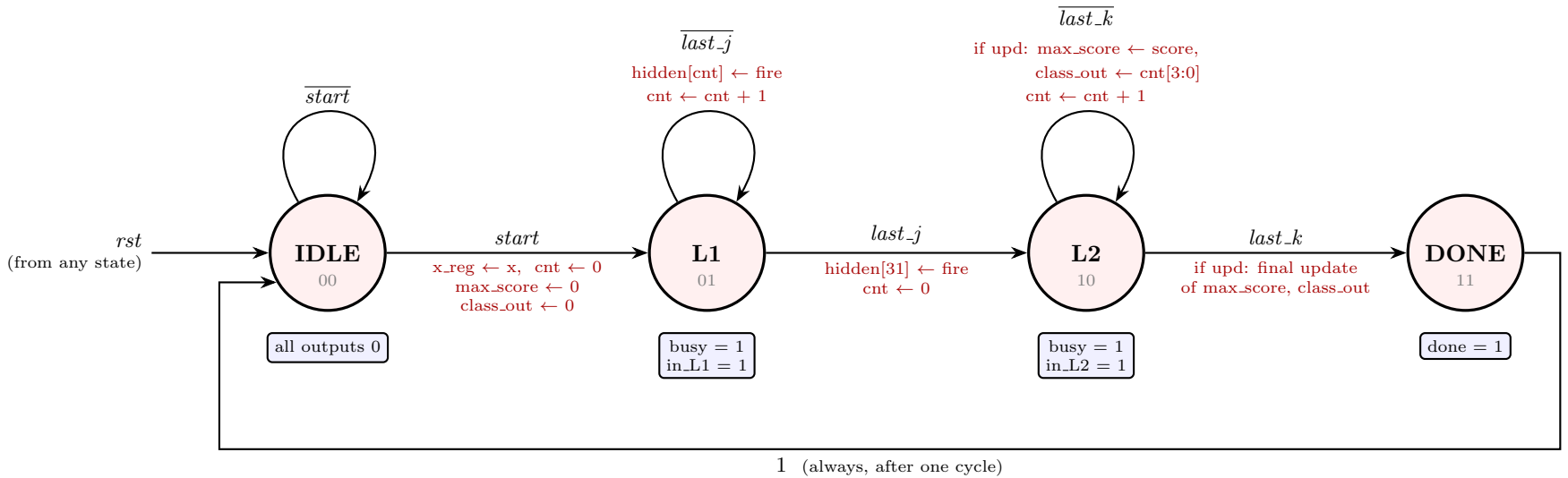
# bnn_top controller, P = 1: state diagram




**Reading this diagram.** Black text on an arrow is the condition checked at the rising edge of `clk`; red text is what the registers load on that same edge. Blue boxes are the Moore outputs of each state (signals not listed are 0). Grey numbers are the 2-bit state codes. Logic: "." = AND, "+" = OR, overbar = NOT. `rst` is synchronous: at any rising edge with `rst` = 1 the state goes to IDLE.

**Signals.** *last_j* = (cnt == 31), *last_k* = (cnt == 9), *upd* = (score > max_score), strict, so a tie keeps the lower class. `load_x` = start . IDLE is the IDLE → L1 edge.

**Timing.** L1 runs 32 cycles (cnt = 0..31) and L2 runs 10 cycles (cnt = 0..9). `done` is high only in DONE, so it is a one-cycle pulse, and the final `class_out` is already loaded when it rises. `hidden` and `class_out` are not written in IDLE or DONE, so they hold until the next `start`. About 44 cycles per image.